

enable cloud communication and safe vehicle control. With the starter kit, engineers have a convenient solution to kickstart their initial evaluation of various automotive applications. It is designed for developing car servers, connected gateways, connectivity modules, and more. Rapid application development becomes achievable with this accessible and feature-rich development board. R-Car starter kits simplify the setup for automotive Linux environments and are ideal for new automotive software engineers. The board features two Ethernet ports, two controller area network flexible data-rate (CAN FD) ports, and one local interconnect network (LIN) port for diverse communication. With 128GiB universal flash storage (UFS) and 64MiB serial peripheral interface (SPI) flash memory, it ensures smooth operations.

Renesas Electronics Corporation
<https://www.renesas.com/in/en>

A development board with Ethernet

The Pico RP2040 board by ADIY is their first microcontroller, featuring a robust dual-core processor, extensive on-chip memory, and a unique programmable I/O subsystem. This provides unmatched capability and versatility for professionals. PICO With Ethernet is a microcontroller evaluation board based on the Raspberry Pi RP2040 and fully hardwired TCP/IP controller W5500—and works the same as the Raspberry Pi Pico board but with additional Ethernet via W5500. The board, tailored for the RP2040, is an affordable, adaptable platform designed to align with the Raspberry Pi Pico's pins. Powered by a dual-core Arm Cortex-M0+ processor, it reaches 133MHz speeds and offers 264kB on-chip SRAM, 4MB Flash memory, and supports 3.3-5.5V inputs. With 27 GPIO pins, four as analogue inputs, it also features a reset button. Its 40nm manufacturing ensures efficiency, mak-



ing it suitable for battery use. The chip, featuring built-in DMA, on-chip libraries, and drag-and-drop programming via USB, streamlines development, and offers enhanced customisability with an on-chip clock, temperature sensor, and 8 I/O state machines.

ADIY
<https://adiy.in>

Graphics cards for professional workloads

AMD has launched two AMD Radeon PRO W7000 series lineup models, the AMD RadeonPRO W7600 and AMD RadeonPRO W7500 workstation graphics processing unit (GPU).



The latest graphics cards are tailored for mainstream tasks in various professional sectors such as media & entertainment, design & manufacturing, architecture, engineering & construction. These graphics cards boost professional processes by prioritising efficiency and enhancing performance. They are fine-tuned to provide top-tier price-to-performance ratios and stability and reliability. The graphics cards offer compute units and ray accelerators at 32 and 28 respectively. The memory interface is set at a 128-bit width. As for the power consumption, the total board power varies: 130W requires a 6-pin connector while 70W can be drawn directly from the peripheral component interconnect express (PCIe) bus. The series is designed to help professionals achieve top-notch results efficiently, even with tight budgets and deadlines.

AMD
<https://www.amd.com>

TEST & MEASUREMENT

A battery health diagnostic tool

KUSAM-MECO has launched the battery internal resistance tester model KM-930. Designed to evaluate the health of rechargeable batteries such as lead storage and lithium types, this

device gauges internal resistance and voltage. It also measures temperature to offer a complete understanding of battery health. The meter employs a 4-terminal test to accurately measure a battery's internal resistance unaffected by contact resistance between the test wire, terminal, and electrode. Controlled by a microprocessor, it precisely detects battery resistance, voltage, and temperature without interrupting the UPS system, utilising AC-low-resistance measurement and noise reduction technology. The device also offers data storage, retrieval, alarm, and auto-shutdown functions, making it suitable for battery production, installation, and equipment manufacturing. The tester model has a USB interface for data transfers and Bluetooth support. Users can store data manually or automatically. Additionally, preset judgement thresholds like PASS, WARNING, and FAIL simplify assessments.

KUSAM-MECO
<https://www.kusamelectrical.com>

Next-generation high-definition oscilloscope

Teledyne LeCroy has launched WaveSurfer 4000HD high-definition oscilloscope with a 30.7cm touchscreen display. The oscilloscope with HD4096 technology has higher resolution over



traditional 8-bit ones and reduced noise for optimal measurements. The 12-bit analogue to digital converters (ADCs)

support the capture of fast signals and support up to 1GHz bandwidth. Furthermore, the company says that a 5GS/s sampling rate guarantees unparalleled accuracy and precision in measurements. The high-performance input amplifiers deliver flawless signal clarity. At the same time, the system's low-noise design ensures an optimal pathway for accurate transmission of signal nuances